

ARRIZA WIBOWO

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SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL

AI / LLM: RAG, prompt engineering, Vercel AI SDK, OpenAI-compatible APIs

Cloud & DevOps: Vercel, Cloudflare, GitHub Actions, Git

Frameworks: Next.js, React, Django, Tailwind CSS, Unity (URP)

Databases: PostgreSQL, Supabase (Auth/RLS/Storage), Firebase

Systems & Security: Active Directory, VMware vCenter/ESXi/vSphere, Clonezilla imaging, Playwright/axe-core, Row-Level Security design

EXPERIENCE

Co-Founder

2026 – Present

Electric Trident LLC, Grand Rapids & Kalamazoo, MI – lead full-stack/AI engineer across client engagements; authored ET-Framework, an internal engineering-standards framework (security review, testing, AI-feature guidelines) adopted company-wide

SwivelServe – B2B E-Commerce Platform (Next.js, Supabase/Postgres, Shopify Admin API, Playwright)

- Architected Shopify as a headless payment/order backend for a real-estate brokerage client, with HMAC-verified webhooks syncing fulfillment through ShipStation.
- Designed a Postgres schema (22 migrations) with row-level security and atomic credit-ledger logic preventing race conditions; authored a 23-spec Playwright e2e suite gating every release.
- Diagnosed and remediated a live production bot-abuse incident by shipping Cloudflare Turnstile and automated cleanup safeguards.

IGNITE Camp AI Platform (Next.js, Vercel AI SDK, Supabase, Drizzle ORM)

- Architected and shipped a production AI-tutoring platform, deployed live and used end-to-end by student teams during an in-person youth innovation camp.
- Built a 3-layer authorization system with passwordless auth for minors, a dual-LLM streaming integration with reasoning-token filtering, and a content-safety system shipped after a real in-camp incident.
- Built a live "prototype engine" streaming sandboxed HTML into an iframe, with a targeted-edit system that cut per-edit token cost ~5.5x versus full regeneration (measured: 83 vs. 456 output tokens).

Liminal Vantage – Expert-Booking Platform (React, Supabase, Stripe, Deno Edge Functions)

- Built a full-stack expert-booking marketplace with 3 auth flows and a 30+ table Postgres schema secured entirely by row-level policies.
- Integrated an LLM-powered recommendation chatbot with prompt-injection defenses across ~31 serverless edge functions handling billing and booking-sync webhooks.

BroncoNav – Accessible Ride-Dispatch Platform (Next.js, Supabase, Playwright/axe-core)

- Built a ride-dispatch platform for a university disability-services office; designed Postgres RLS as the authorization boundary and audited/fixed two privilege-escalation vulnerabilities.
- Built an automated WCAG 2.1 AA accessibility test harness (Playwright + axe-core) covering 50 success criteria, feeding a formal VPAT/ACR compliance submission.

Research Programmer

June 2026 – Present

Dr. Lina Sawalha Research Team, WMU ECE Dept. – NSF-funded mobile GPU energy study (Unity, C#, Python, ADB); sole developer, Independent Study ECE 4900

- Diagnosed and fixed 15 defects in an inherited Unity benchmark that had never run successfully, delivering a fully automated, sentinel-verified Android test harness.
- Designed a JSON-driven config system generating 48 graphics-test configurations across 15 rendering parameters, and ported the benchmark from Unity 2020.3 to Unity 6.3.
- Built a Python orchestration and analysis toolchain from scratch for automated sweep execution and results reporting.
- Preparing to integrate on-device energy profiling via Gator (Gatord performance counters) and Perfetto power-rail capture for the study's upcoming measurement phase.

Research Programmer

Sept 2025 – Jan 2026

Dr. Hexu Liu Research Team, WMU

AI Revit Copilot – Revit Automation Extension (Revit API, LLM Integration, RAG, Python)

- Designed a modular system translating natural-language commands into safe, validated Revit API actions, with context extraction for real-time model state awareness.
- Refactored the extension from modal to modeless via the ExternalEvent pattern, and implemented rule-based 3D model generation from a strict keyword/parameter rules file derived from the Revit documentation.

System Administrator

January 2025 – Current

Western Michigan University College of Engineering, Kalamazoo, MI

- Migrated college-owned devices to Active Directory, enabling secure role-based access for students, faculty, and staff.
- Manage virtualized infrastructure for engineering software and license servers via VMware vCenter and ESXi, using vSphere for remote administration; migrated the CATIA license server from a VM to a dedicated physical host.
- Built and maintain a centralized file-sharing server for the College of Engineering, giving labs and faculty secure, organized access to shared resources.
- Create and deploy standardized PC/laptop images with Clonezilla; design specialized images for faculty research environments.

EDUCATION

Bachelor of Science in Computer Science

Expected April 2028

Western Michigan University, Kalamazoo, MI – Global Education Undergraduate Merit Scholarship (\$10,000/yr, up to 4 years) · 4.0 GPA